Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	20 October 2026
Team ID	En25cs3010862
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, SQL HTML, CSS, etc	Used for data cleaning, calculating risk scores, statistical summaries, and querying health record datasets.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	Snowflake / MySQL / Excel / CSV	Stores the 5,000 patient health records containing demographic vitals, lifestyle habits, and chronic disease flags
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stages / COPY INTO / Tableau Data Connections	Efficiently loads raw patient telemetry and clinical CSV exports into the analytics workspace.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Eg. Tableau prep, Snowflake SQL, Tableau Desktop	Computes health risk scores, bins age cohorts, and standardizes BMI, blood pressure, and glucose metrics.

5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Snowflake SQl Charts, Tableau , Python	Delivers interactive visual insights on diabetes, stroke, and heart disease prevalence across demographics.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask,Streamlit	Provides clinicians, analysts, and patients with an intuitive dashboard to filter and explore chronic risk factors.